A. Sharper bounds for the finite profiles

The norm choices in (1) and (4) are sufficient choices for the general parameter family. For the finite profiles we use the following sharper bounds for the same distributions and norm tests. They replace the conservative numerical sufficient conditions, without changing KeyGen or the security games.

Write $\rho_t(c) = \sum_{z\in\mathbb{Z}} \exp(-(z-c)^2/(2t^2))$ and $\rho_t = \rho_t(0)$. Poisson summation gives $\rho_t(c) \le \rho_t$ and, for $t' \ge t$, $\rho_{t'}/\rho_t \le t'/t$. The first inequality follows by bounding each cosine in the dual series by one. For the second, the dual series $\rho_t/(\sqrt{2\pi}t) = \sum_{k\in\mathbb{Z}} \exp(-2\pi^2 t^2 k^2)$ is decreasing in t.

Suppose $\Pr[f \in \mathcal{R}_q^\times] \ge \tau > 0$. For any $0 < u < 1$, put $t = u/(8\sigma^2)$. Then the key rejection probability obeys

$$\eta_k \le \tau^{-1} \exp\left(-tB_k^2 + \frac{t}{1-u}\right) \left(\frac{\rho_{\sigma/\sqrt{1-u}}}{\rho_\sigma}\right)^n \cdot \left(\frac{\rho_{\sigma/\sqrt{1-u/4}}}{\rho_\sigma}\right)^{(D-1)n}. \tag{1}$$

To see this, let $X \leftarrow D_{\mathbb{Z},\sigma}$. Completing the square gives

$$-\frac{X^2}{2\sigma^2} + t(2X+1)^2 = -\frac{1-u}{2\sigma^2}\left(X - \frac{u}{2(1-u)}\right)^2 + \frac{t}{1-u}.$$

Only the constant coefficient of $f = 2f_0 + 1$ has the shift by one. Its exponential moment is therefore at most $\exp(t/(1-u))\rho_{\sigma/\sqrt{1-u}}/\rho_\sigma$. Each remaining coefficient of f contributes the same ratio without that exponential factor; each of the other $(D-1)n$ coefficients contributes $\rho_{\sigma/\sqrt{1-u/4}}/\rho_\sigma$. Markov's inequality at B_k^2 and division by the invertibility lower bound τ prove (7).

For any $0 < v < 1$, the same exponential-moment argument and the second theta inequality give

$$1 - p_B \le \exp\left(-\frac{vB_s^2}{2r^2}\right)(1-v)^{-Dn/2}. \tag{2}$$

In particular this is an integer-Gaussian bound, rather than substitution of a continuous chi-square probability.

For the three profiles $q \equiv 1 \pmod{2n}$ and $\rho_\sigma < 2$. Let $\zeta \in \mathbb{F}_q$ have order $2n$, let $H = \langle\zeta\rangle$, write $[x]_q$ for the centered residue of x, and set

$$E_{n,q} = \min_{a\in\mathbb{F}_q^\times} \frac{1}{2q^2} \sum_{x\in aH} [x]_q^2, \qquad p_1 = \frac{\exp(-1/(2\sigma^2))}{\rho_\sigma}.$$

Then an admissible invertibility lower bound is

$$\tau = 1 - \frac{n}{q} - n\left(1 - \frac{1}{q}\right)\exp(-16p_1E_{n,q}). \tag{3}$$

Indeed the characteristic function $\phi(s) = \mathbb{E}\exp(2\pi i s X)$ is real, and $\phi(s) \ge 2\Pr[X = 0] - 1 > 0$. Retaining the $X = \pm 1$ terms gives

$$0 \le \phi(s) \le 1 - 2p_1(1-\cos(2\pi s)) \le \exp(-16p_1 \operatorname{dist}(s,\mathbb{Z})^2).$$

For a root $\xi = \zeta^{2j+1}$ of $X^n + 1$, finite Fourier inversion therefore bounds $\Pr[f(\xi) = 0]$ by $q^{-1} + (1 - q^{-1})\exp(-16p_1E_{n,q})$. Here $\{\pm\xi^k : 0 \le k < n\} = H$ because n is a power of two. Thus the sum of the n squared distances in each nontrivial Fourier product is at least $E_{n,q}$; multiplication by two only permutes the nonzero Fourier frequencies. A union bound over all n roots proves (9).

Exact coset enumeration yields

q	ζ	$E_{n,q}$
12289	10302	11932987670/302039042
15361	4989	18916242006/471920642

The fixed (u, v) values used for profiles 1–3, interpreted as exact decimal rationals, are

	u	v
1	0.6070989879390658	0.3810220434537782
2	0.5902687398832	0.36167515955237883
3	0.6768302726631384	0.35746618525530693

The outward-rounded supplementary calculation checks $\eta_k \le 2^{-\lambda}$, $1 - p_B \le 2^{-\lambda}$, $LM^{-d}(1-2^{-\lambda}) \ge \lambda^2$, and $4U\exp(-\tau(1-2^{-\lambda})L_k) \le 2^{-\lambda}$ for the three profiles. This establishes the stated norm and cutoff inequalities for these distributions.